

SOME POPULAR PROCESSORS

Name	Features
Intel Xeon E7-8894 v4	2.4GHz with turbo boost to 3.4GHz with 60MB cache
Intel i7-7700	3.6GHz with turbo boost to 4.2GHz with 8MB cache
AMD FX series	2.8GHz to 4.5GHz with up to 8MB L3 cache
AMD Ryzen 7 1800X	3.6GHz base clock and 4GHz boost clock
Microchip ATtiny1617	16KB flash; sleep currents up to 100nA
NXP i.MX6ULL	528MHz with 128KB L2 cache
NXP LS1028A	64-bit ARM Cortex V8 processor core
Omni Vision OV788& OV798	32-bit RISC MCUs with 32KB data cache and 32KB instruction cache
Analog Devices ADuCM4050	40µA/MHz with 680nA in hibernate mode

from Intel that allows clock speed of the processor to be dynamically changed by the software. It allows the processor to meet the instantaneous performance needs of the operation being performed, thus minimising power consumption and loss through heat.

LongHaul is a speed-throttling technology developed by VIA Technologies that helps reduce power consumption. It uses specialised instructions utilised by the software to exercise control on bus-to-core frequency ratio and CPU core voltage. When the system boots up, ratio and voltage are set to hardware defaults. Depending on the load on the CPU, a CPU driver controls the throttling during operation.

Following a similar course, AMD's dynamic frequency scaling and power saving known as PowerNow (mobile chips) and Cool'n'Quiet (desktop and server chips) allowed adjusting clock speed

under varying loads and idle conditions to save battery power. These power-saving techniques allowed power saving but, caused problems with the operating systems. Microsoft also reported problems previewing video files with SpeedStep enabled under Windows 2000 or XP. This called for making further changes, resulting in newer protocols that are being implemented today.

Power saving for the present.

Building up on PowerNow, AMD launched Turbo Core beginning with Bulldozer architecture. The architecture focused on reducing power consumption over the earlier K10 architecture for higher performance per watt.

Bulldozer processors are based on GlobalFoundries' 32nm silicon on insulator, which reuses DEC approach for multitasking computer performance. By eliminating redundant elements that creep into multi-core designs, it aims to take advantage of its hardware capabilities while using less power.

This is followed by Zen architecture that looks to define balance between power and performance. As per multi-threaded Cinebench R15, performance per watt is more than double than it was for Piledriver architecture.

Zen architecture includes a complex stack engine that can eliminate certain stack manipulation instructions. This coupled with the

ARM Cortex®-M4

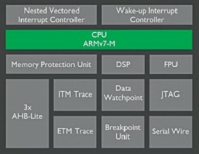


Fig. 2: Cortex M4 has been very popular with low-power processors for the IoT

core design, which is optimised to use less power, is something to look out for. It has an integrated power management controller that monitors and adjusts the voltage used by each core according to the temperature and loading.

Unlike previous versions, power management roles are baked directly into silicon and are out of the control of the software and operating system. The hardware on the chip can respond in a matter of milliseconds, allowing much tighter control over voltages and clock speeds.

Balancing power and performance. Intel Turbo Boost draws a parallel with Turbo Core from AMD. However, Enhanced Intel SpeedStep Technology is more power conscious and allows the system to dynamically adjust processor voltage and core frequency, decreasing average power consumption and heat production.

Design strategies in SpeedStep involve separation between voltage and frequency changes. Varying voltages in small increments separately from frequency changes allow the processor to reduce periods of system unavailability occurring during frequency changes. The system can transition between voltage and frequency states more often, hence providing improved power-performance balance.

Clock partitioning and recovery is another method for reducing power

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